

Signed Lane-Exit Graphs and Complete Survivor Frontiers

Phase Descent in Full Ordinary-Lane Five-Token Dynamics

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2026

This paper (Paper XXXVI of the RIME program) studies intrinsic raw survivor incidence in a declared multi-lane regime. It does not assert typed transfer membership, projectability, recursive compatibility, credit settlement, or a reset bound.

Abstract

Problem. A spectator-safe fused-pair witness does not determine the placements of the three surviving source lineages. In multiple arithmetic lanes, a compressed mobility graph must also support a single concrete guarded path through its internal transitions and terminal exit.

Approach. The analysis uses normalized returns $\phi_r^{(a)} = \varepsilon_\Delta p^r a$ with $n = 5g$, $\Delta = g$, and $g \geq 2$. The source fills an ordinary five-point lane. The branch stabilizes the lane partition and restricts to a cycle-graph isomorphism between ordinary lanes; its action on the four-point punctured lane is unrestricted. Full ordinary lanes are invariant under guarded returns. Their edge guards are uniform over all source-labelled representatives, and globally enabled return blocks realize every within-lane phase.

Results. Quotienting reachable injections by phase gives exactly the reachable internal vertices of a signed lane graph. Its state is a lane together with source-relative orientation. Each boundary edge can be realized, on a genuinely reachable phase representative, with any prescribed source-consecutive pair at the terminal kernel. Consequently, a consecutive pair has precisely one survivor injection for each reachable exit sign: the three survivors occupy $(2\Delta, 3\Delta, 4\Delta)$ in source order for positive sign, and $(4\Delta, 3\Delta, 2\Delta)$ for negative sign. Nonconsecutive pairs have no strict-exit witness. An all- g matched family has identical unsigned labelled lane graphs and fused-pair spectra but different survivor spectra.

Boundary. Phase can be removed from the unbudgeted reachability state, whereas unsigned lane data cannot determine the complete raw frontier. This is a scoped sufficiency and non-descent theorem, not a classification of the full lane stabilizer or a universal minimality assertion.

Keywords. circular automata; single-defect dynamics; packet lineages; signed graphs; phase quotient; guarded reachability; survivor incidence

Notation Table

Symbol	Meaning
$Q = \mathbb{Z}/(5g)\mathbb{Z}, \Delta = g$	labelled circular carrier and lane spacing
$p(q) = q + 1$	fixed full cycle
$E = Q \setminus \{0\}$	normalized section carrier
ε_Δ	collapse $0 \mapsto \Delta$, fixing every other point
$K_\Delta = \{0, \Delta\}$	terminal collision pair
C_*	four-point punctured lane
$C^{(j)}, \mathcal{C}_\Delta^{(5)}$	ordinary lanes and their collection
$\widehat{C}_*$	full five-point pre-collapse kernel lane
ρ_C	positive Δ -rotation of a five-point lane
a	lane-stabilizing branch, locally dihedral on ordinary lanes
$\phi_r^{(a)}$	normalized guarded return of label r
$T_0 = \{q_0, \dots, q_4\}$	fixed source-lineage labels in positive source order
$\lambda_0, C_{\text{src}}$	source injection and its ordinary lane
$\mathcal{L}_a(\lambda_0)$	injections reachable by actual guarded paths
$\sim_{\text{ph}}$	equivalence by within-lane phase rotation
$\widetilde{\Gamma}_{\text{ori}}^{\text{ph}}$	signed lift of the lane-exit graph
$\text{ExitOri}_a(C_{\text{src}})$	reachable boundary signs
$\mathcal{W}_a(\lambda_0; F, \mu)$	terminal witness fiber
$\mathfrak{S}_a(\lambda_0; F)$	source-addressed survivor spectrum

1 Introduction

A raw strict exit records more than the pair that fuses. It also records the coordinates of every surviving source lineage. Forgetting those placements can make distinct strict-exit frontiers indistinguishable at the fused-pair level. Paper XXXIV establishes this separation in a one-lane dihedral family [5].

Paper XXXV goes further within one lane: for arbitrary branch permutations, reachable cyclic orders determine the complete raw survivor frontier and terminal phases [3]. The present paper instead addresses mobility between ordinary lanes, where phase compression must preserve a single concrete witness along the entire path.

The multi-lane problem adds a witness issue. An existential transition between two coarse classes cannot generally be composed with another such transition: the outgoing edge may require a representative different from the one actually reached. Paper XXXIII therefore retains source-addressed intermediate incidence rather than an unrestricted Boolean composition of orbit edges [4]. A positive compression theorem must discharge this representative mismatch, not merely hide it.

This paper gives such a theorem in a capacity-isolated regime. Fix $n = 5g$, $\Delta = g$, and a source occupying an entire ordinary five-point lane. The punctured lane has only four points, so it cannot receive the complete source under an injective return. Assume that each ordinary-lane restriction of the branch is a rotation or reflection. Then every internal lane edge is enabled on its entire representative fiber, and its orientation law is uniform. A finite-order return block supplies all five phases by actual guarded paths.

The correct compressed state is not the lane alone, but the lane together with source-relative orientation. The paper proves three statements.

1. Reachable injections modulo phase are exactly reachable signed lane vertices, with uniform edgewise lifting from the same representative.
2. Reachable boundary signs determine the complete source-addressed survivor frontier, including the converse construction of every claimed incidence.

3. Unsigned labelled graphs do not suffice: an explicit all- g pair of branches has the same unsigned graph and pair spectrum, but one versus two survivor injections for every consecutive source pair.

The argument is data-independent. The normalized action law is inherited from the branch-coordinate framework [2, 6], but all geometric and witness-lifting statements used here are proved below.

1.1 Related Work and Novelty Boundary

Order-preserving automata and their synchronization algorithms are classical. Eppstein studies reset sequences for automata whose transitions respect a cyclic ordering [8]. Circular automata also belong to the established reset-bound literature [7]. The question here is instead the unbudgeted first-exit incidence of five fixed source lineages: every internal step remains injective on their complete support, and the terminal event identifies exactly a prescribed source pair.

Almost-group automata with one defect-one letter and permutation letters are a closely related ambient class. Casas Torres studies complete reachability, including transitive imprimitive permutation groups [1]. Complete reachability concerns support images; the present theorem retains source-labelled orientation and all three survivor coordinates on one guarded witness. No complete-reachability characterization is claimed.

Synchronization under dynamic constraints studies constraints on the order in which words traverse states [9]. Here the constraint is injectivity on the five-point occupied support at every internal return, followed by a specified source-marked collision. The signed graph is an exact quotient proved for this declared full-lane regime, not a general reduction for constrained synchronization.

Within the RIME line, Paper XXX supplies universal section symmetries; Paper XXXI normalizes partial returns; Paper XXXIII identifies a four-versus-five capacity obstruction; Paper XXXIV separates pair and survivor observables; and Paper XXXV completes the arbitrary-branch one-lane raw frontier through cyclic-order reachability and terminal-phase collapse [2–6]. The contribution here is the multi-lane local-dihedral signed quotient, uniform same-witness path lifting, a complete survivor theorem, and a matched symbolic obstruction to unsigned descent. The proofs do not assume a typed recursive-return theorem.

2 Normalized System and Full-Lane Regime

2.1 Action and Reachability

Throughout,

$$n = 5g, \quad \Delta = g, \quad g \geq 2, \quad Q = \mathbb{Z}/n\mathbb{Z}, \quad E = Q \setminus \{0\}. \quad (2.1)$$

Set $p(q) = q + 1$ and

$$\varepsilon_{\Delta}(q) = \begin{cases} \Delta, & q = 0, \\ q, & q \neq 0. \end{cases} \quad K_{\Delta} = \{0, \Delta\}. \quad (2.2)$$

The Δ -lane partition of E consists of

$$\begin{aligned} C_* &= \{\Delta, 2\Delta, 3\Delta, 4\Delta\}, \\ C^{(j)} &= \{j + t\Delta : t \in \mathbb{Z}/5\mathbb{Z}\}, \quad 1 \leq j \leq g-1, \\ \mathcal{C}_{\Delta}^{(5)} &= \{C^{(1)}, \dots, C^{(g-1)}\}, \\ \widehat{C}_* &= \{0, \Delta, 2\Delta, 3\Delta, 4\Delta\}. \end{aligned} \quad (2.3)$$

On every ordinary lane and on $\widehat{C}_*$, positive rotation is $\rho_C(q) = q + \Delta$.
 Fix a permutation a of E stabilizing this partition. For each ordinary lane, require

$$a(C) = C', \quad a\rho_C = \rho_{C'}^{\chi_C} a \quad \text{on } C, \quad \chi_C \in \{+1, -1\}. \quad (2.4)$$

Thus $a|_C$ is a five-cycle graph isomorphism. Its restriction to C_* is arbitrary. This is the **full ordinary-lane local-dihedral regime**; neither normalizer membership nor a common multiplier is assumed.

The return of label $r \in Q$ is

$$\phi_r^{(a)} = \varepsilon_\Delta \circ p^r \circ a. \quad (2.5)$$

Composition is right-to-left: a label first applies a , then p^r , then ε_Δ . A path with labels $r_1, \dots, r_k$ applies them in that listed order. Labels are normalized return labels, not individual original-automaton letters.

Let T_0 be five fixed source labels, and fix an injection $\lambda_0 : T_0 \hookrightarrow E$ whose image is an ordinary lane C_{src} . Write $q_0, \dots, q_4$ in the positive cyclic order induced by λ_0 ; source subscripts are modulo 5.

A return from λ is enabled exactly when $\phi_r^{(a)}$ is injective on $\lambda(T_0)$. Equivalently,

$$K_\Delta \not\subseteq p^r a(\lambda(T_0)). \quad (2.6)$$

Write $\mathcal{L}_a(\lambda_0)$ for the injections reachable by finite enabled paths, including the empty path. No length budget is imposed.

2.2 Terminal Witnesses

Fix $F \subset T_0$ of size two and $R_0 = T_0 \setminus F$. A formal pre-collapse placement is an injection $\mu : T_0 \hookrightarrow Q$ with $\mu(F) = K_\Delta$. Define

$$\mathcal{W}_a(\lambda_0; F, \mu) = \left\{ (u, \lambda) : \begin{array}{l} u \in Q \setminus \mu(T_0), \\ \lambda \in \mathcal{L}_a(\lambda_0), \\ \lambda = a^{-1} p^{-u} \mu \end{array} \right\}. \quad (2.7)$$

The hole condition makes the pullback well typed in E . Its projection onto u is the phase fiber $\Phi_a(\lambda_0; F, \mu)$.

Proposition 2.1 (exact terminal pullback).

A formal placement μ is attainable by a guarded prefix and one terminal strict return if and only if $\mathcal{W}_a(\lambda_0; F, \mu) \neq \emptyset$. For fixed μ , projection onto u is a bijection from $\mathcal{W}_a$ to Φ_a .

Proof. An actual terminal call has $\mu = p^u a \lambda$ with λ reachable. Since $p^{-u} \mu(T_0) = a(\lambda(T_0)) \subseteq E$, its terminal hole satisfies $u \notin \mu(T_0)$. Inverting the two bijections gives the pullback in (2.7). Conversely, every pair in (2.7) gives $\mu = p^u a \lambda$ after an actual guarded prefix. The injection μ puts precisely the two labels of F at $0, \Delta$, so the final collapse has rank four. The pullback is unique for fixed (μ, u) , proving the projection assertion.

The resulting strict incidence is

$$\beta_\mu(q) = \begin{cases} \Delta, & q \in F, \\ \mu(q), & q \in R_0. \end{cases} \quad (2.8)$$

Define

$$\begin{aligned} \mathfrak{S}_a(\lambda_0; F) &= \{\mu|_{R_0} : \mu(F) = K_\Delta, \mathcal{W}_a(\lambda_0; F, \mu) \neq \emptyset\}, \\ \mathfrak{F}_a(\lambda_0) &= \{F : |F| = 2, \mathfrak{S}_a(\lambda_0; F) \neq \emptyset\}. \end{aligned} \quad (2.9)$$

If fixed nonempty disjoint source blocks are supplied, (2.8) reconstructs the packet endpoint by placing each surviving block at its own coordinate and the union of the two fused blocks at Δ . No block mass, role, or ancestry field enters the present dynamics.

3 Signed Phase Descent

3.1 Uniform Edge Laws and Actual Phase Control

Lemma 3.1 (full-lane invariance and uniform lifting).

Suppose $a(C^{(j)}) = C^{(j')}$ and let $h \equiv j' + r \pmod{g}$, with $0 \leq h \leq g - 1$. The label r is disabled on every injection filling $C^{(j)}$ when $h = 0$. When $h \neq 0$, it is enabled on every such injection, maps it onto $C^{(h)}$, and has sign $\chi_{C^{(j)}}$.

For the enabled restriction $f : C \rightarrow C'$, every phase k satisfies

$$f\rho_C^k = \rho_{C'}^{\chi_{C^{(j)}}k} f. \quad (3.1)$$

Proof. If $h = 0$, the full set $p^r a(C^{(j)})$ is $\widehat{C}_*$. It contains the two collision points, so the collapse is not injective. If $h \neq 0$, this full set is $C^{(h)}$, which avoids 0. The collapse is the identity there, and the restriction is $p^r a|_{C^{(j)}}$. Translation preserves positive Δ -orientation, so (2.4) gives its sign and (3.1). The guard reads only the full support, not the representative filling it. Composition with any bijection $T_0 \rightarrow C$ therefore gives a legal injection onto C' .

In particular, every reachable source state still fills an ordinary lane. The four-point punctured lane is never an internal destination.

Lemma 3.2 (actual phase saturation).

If λ is reachable and fills C , every $\rho_C^k \lambda$ is reachable.

Proof. Let $m = \text{ord}(a)$ and put $\psi_\Delta = \varepsilon_\Delta p^\Delta|_E$. This is a permutation of E : $p^\Delta(E)$ omits Δ , and the collapse replaces its point 0 by that missing point. Thus

$$\phi_0^{(a)} = a, \quad \phi_\Delta^{(a)} = \psi_\Delta a \quad (3.2)$$

are globally enabled on injections into E . The actual label block consisting of $m - 1$ zeros followed by Δ implements

$$\phi_\Delta^{(a)} (\phi_0^{(a)})^{m-1} = \psi_\Delta a^m = \psi_\Delta. \quad (3.3)$$

Every intermediate step is enabled. On an ordinary lane, $\psi_\Delta|_C = \rho_C$. Repeating this block realizes all five phases. This proves saturation by actual paths, not by a formal symmetry identification.

3.2 The Signed Graph

The base graph Γ_a^{ori} has ordinary-lane vertices and a boundary vertex ∂ . Its internal and boundary edges are

$$\begin{aligned} C &\xrightarrow{r, \chi} C' &\iff \phi_r^{(a)}|_C : C \rightarrow C' \text{ is bijective of sign } \chi, \\ C &\xrightarrow{u, \chi_\partial} \partial &\iff p^u a(C) = \widehat{C}_* \text{ with pre-collapse sign } \chi_\partial. \end{aligned} \quad (3.4)$$

These are different edge types: a boundary exponent ends the path with a strict collision, not another enabled internal return.

The **signed lift** $\tilde{\Gamma}_a^{\text{ori}}$ has internal vertices (C, σ) and boundary vertices (∂, σ) , with $\sigma \in \{+1, -1\}$. It lifts the edges as

$$\begin{aligned} (C, \sigma) &\xrightarrow{r} (C', \sigma\chi), \\ (C, \sigma) &\xrightarrow{u} (\partial, \sigma\chi_\partial). \end{aligned} \tag{3.5}$$

Set

$$\text{ExitOri}_a(C_{\text{src}}) = \{\chi : (C_{\text{src}}, +1) \rightsquigarrow (\partial, \chi) \text{ in } \tilde{\Gamma}_a^{\text{ori}}\}. \tag{3.6}$$

The source-relative orientation of a reachable injection is positive when its cyclic label order is $(q_0, \dots, q_4)$, and negative when that order is reversed. Lemma 3.1 shows that these are the only two possible orders. For a fixed base point $c_C \in C$, such an injection has the form

$$\lambda(q_i) = \rho_C^{b+\sigma i}(c_C), \quad b \in \mathbb{Z}/5\mathbb{Z}, \quad \sigma \in \{+1, -1\}. \tag{3.7}$$

Define $\lambda \sim_{\text{ph}} \lambda'$ when they have the same lane and $\lambda' = \rho_C^k \lambda$ for some k .

Theorem 3.3 (signed phase descent).

The map

$$[\lambda]_{\text{ph}} \mapsto (\lambda(T_0), \text{sgn}_{\lambda_0}(\lambda)) \tag{3.8}$$

is a bijection

$$\mathcal{L}_a(\lambda_0)/\sim_{\text{ph}} \cong \text{Reach}_{\tilde{\Gamma}_a^{\text{ori}}}^{\text{int}}(C_{\text{src}}, +1). \tag{3.9}$$

It is an isomorphism of the reachable phase-quotient transition graph with the reachable internal signed graph, including return labels and disabled labels. Every finite internal signed path lifts, edge by edge, from the same concrete initial representative.

Proof. Project an actual guarded path. Lemma 3.1 keeps its support in ordinary lanes and updates its source sign exactly as (3.5). This gives a signed path and proves that the map in (3.8) takes values in the right-hand side.

Conversely, let a signed path have labels $r_1, \dots, r_k$. Starting with the given λ_0 , define recursively $\lambda_i = \phi_{r_i}^{(a)} \lambda_{i-1}$. Each edge is enabled on every injection filling its incoming lane by Lemma 3.1. In particular it is enabled on the representative just produced by the previous edge. Induction supplies one actual path with the required support and sign. This proves surjectivity without choosing independent edge witnesses.

Two reachable injections with the same lane and sign have the same cyclic label order and differ only by b in (3.7), hence are phase equivalent. Opposite signs cannot be phase equivalent for five distinct labels. Lemma 3.2 supplies every representative of each reachable phase class. Finally, enabledness is support-dependent, and (3.1) makes the output phase class independent of the incoming representative. These facts prove the transition-graph assertion.

The quotient has at most $2(g-1)$ internal vertices and the reachable injection set has at most $10(g-1)$ elements. This does not preserve the cost of reaching a particular phase: phase control may insert the blocks in (3.3). No shortest-word or fixed-endpoint word claim follows.

4 Boundary Realization and Complete Survivor Frontier

4.1 Phase-Complete Boundary Lifting

Lemma 4.1 (prescribed-pair boundary realization).

Let (C, σ) be reachable, let $b = p^u a|_C : C \rightarrow \widehat{C}_*$ be a fixed boundary edge of sign χ_∂ , and fix a source-consecutive pair $F = \{q_i, q_{i+1}\}$. Within any reachable incoming phase fiber, exactly one phase puts F at K_Δ through this same exponent u . The resulting terminal witness has sign $\sigma\chi_\partial$.

Proof. Choose a reachable incoming λ from Theorem 3.3. Every $\lambda^{(k)} = \rho_C^k \lambda$ is reachable by Lemma 3.2. Writing $\hat{\rho}(q) = q + \Delta$ on $\widehat{C}_*$, we have

$$\mu^{(k)} = b\lambda^{(k)} = \hat{\rho}^{\chi_{\partial^k}}(b\lambda). \quad (4.1)$$

All five placements have the same sign $\sigma\chi_{\partial}$. The image of a source-consecutive pair is an edge of the target five-cycle. Its five rotations are distinct and exhaust the target edges. Exactly one is K_{Δ} , which selects a unique k_F .

Put $\lambda_F = \lambda^{(k_F)}$ and $\mu_F = p^u a \lambda_F$. The same placement verifies the fused pair, orientation, and all survivor coordinates. Moreover, $p^{-u} \mu_F(T_0) = a(C) \subseteq E$, so $u \notin \mu_F(T_0)$. Hence (u, λ_F) belongs to the witness fiber (2.7). Concatenating the actual incoming path, the phase-control block, and the terminal return gives a single complete witness.

This uniqueness is for the chosen phase fiber and chosen boundary exponent; it does not assert uniqueness of all paths or all terminal witnesses.

4.2 Terminal Rigidity

For a sign χ , let $\mathfrak{S}_{\Delta}^{\chi}(F)$ be the survivor restrictions of bijections $\mu : T_0 \rightarrow \widehat{C}_*$ such that $\mu(F) = K_{\Delta}$ and μ has source-relative sign χ .

Lemma 4.2 (terminal rigidity).

A nonconsecutive pair has no such placement. For $F = \{q_i, q_{i+1}\}$ there is exactly one placement of each sign:

$$\begin{aligned} \mu_F^+(q_{i+t}) &= t\Delta, \\ \mu_F^-(q_{i+t}) &= (1-t)\Delta, \quad t \in \mathbb{Z}/5\mathbb{Z}. \end{aligned} \quad (4.2)$$

Consequently, on the ordered survivor labels $(q_{i+2}, q_{i+3}, q_{i+4})$,

$$s_F^+ = (2\Delta, 3\Delta, 4\Delta), \quad s_F^- = (4\Delta, 3\Delta, 2\Delta). \quad (4.3)$$

In particular, the two survivor injections are distinct.

Proof. Preserving or reversing the source order makes μ a five-cycle graph isomorphism. The preimage of the target edge $\{0, \Delta\}$ is therefore a source edge. For a consecutive F , positive sign forces $q_i \mapsto 0$, $q_{i+1} \mapsto \Delta$; negative sign forces the reverse assignment. The rest of the cycle then forces (4.2). Restricting gives (4.3), whose first coordinates differ.

4.3 Completion Theorem

Theorem 4.3 (complete survivor frontier).

In the declared regime,

$$\mathfrak{S}_a(\lambda_0; F) = \begin{cases} \bigsqcup_{\chi \in \text{ExitOri}_a(C_{\text{src}})} \mathfrak{S}_{\Delta}^{\chi}(F), & F \text{ source-consecutive,} \\ \emptyset, & F \text{ nonconsecutive.} \end{cases} \quad (4.4)$$

The union is a disjoint union of actual survivor injections, not merely of witnesses carrying different sign labels.

Proof. For the forward inclusion, choose an actual terminal witness (u, λ) and put $\mu = p^u a \lambda$. The reachable injection fills an ordinary lane C by Lemma 3.1. Its translated image is a full residue class modulo g . Since $\mu(F)$ contains 0, that class is $\widehat{C}_*$. Thus u defines a boundary edge from C .

The actual prefix projects by Theorem 3.3 to a signed path ending at (C, σ) . Appending this boundary edge gives exit sign $\chi = \sigma\chi_{\partial}$. The very same μ has this sign, puts F at the kernel, and

supplies the survivor restriction. Lemma 4.2 forces F to be consecutive and places its survivors in $\mathfrak{S}_\Delta^\chi(F)$.

For the reverse inclusion, choose $\chi \in \text{ExitOri}_a(C_{\text{src}})$ and a formal placement μ contributing to $\mathfrak{S}_\Delta^\chi(F)$. Take a signed path to this boundary vertex. Lift its internal part using Theorem 3.3, and retain its fixed final boundary exponent u . Lemma 4.1 then gives an actual reachable phase representative and a placement ν with the prescribed pair and sign χ . Both ν and μ fill $\widehat{C}_*$ and have that same pair and sign. Lemma 4.2 implies $\nu = \mu$. Thus the claimed survivor injection has an actual witness, not only a formal terminal shape. Finally, (4.3) proves disjointness.

Corollary 4.4 (explicit incidence spectrum).

The fused-pair spectrum is exactly

$$\mathfrak{F}_a(\lambda_0) = \{\{q_i, q_{i+1}\} : i \in \mathbb{Z}/5\mathbb{Z}\}. \quad (4.5)$$

For each such F , its survivor spectrum consists of s_F^+ if positive exit sign is reachable, s_F^- if negative exit sign is reachable, and both if both are reachable. In particular,

$$|\mathfrak{S}_a(\lambda_0; F)| = |\text{ExitOri}_a(C_{\text{src}})| \in \{1, 2\}. \quad (4.6)$$

Proof. A boundary exponent exists directly from C_{src} : translate its image under a to residue zero modulo g . Hence the exit-sign set is nonempty. Apply Theorem 4.3 and Lemma 4.2.

Writing $\beta_F^\chi = \varepsilon_\Delta \mu_F^\chi$, the complete strict-incidence spectrum is

$$\begin{aligned} \mathfrak{B}_a(\lambda_0) &= \{\beta_F^\chi : F \text{ consecutive, } \chi \in \text{ExitOri}_a(C_{\text{src}})\}, \\ |\mathfrak{B}_a(\lambda_0)| &= 5|\text{ExitOri}_a(C_{\text{src}})| \in \{5, 10\}. \end{aligned} \quad (4.7)$$

Indeed, $\beta^{-1}(\Delta)$ recovers its fused source pair and (4.3) distinguishes the two signs for that pair. All incidences have output support $\{\Delta, 2\Delta, 3\Delta, 4\Delta\}$. Their source-addressed differences are not visible from that common support.

5 Unsigned Lane Data Do Not Descend

The unsigned graph retains lanes, internal return labels and guards, boundary exponents, and the source lane, but forgets every orientation sign. The following obstruction preserves this full unsigned data, not merely unlabelled connectivity.

Theorem 5.1 (all- g unsigned non-descent).

For every $g \geq 2$, there are two branches in the declared regime which, for the same source injection, have identical unsigned labelled lane-exit graphs and identical fused-pair spectra, but different survivor spectra for every consecutive source pair.

Proof. Let $a_+ = \text{id}_E$. Define a_- to fix C_* pointwise and reflect every ordinary lane:

$$a_-(j + t\Delta) = j - t\Delta, \quad 1 \leq j \leq g-1, \quad t \in \mathbb{Z}/5\mathbb{Z}. \quad (5.1)$$

Both branches fix each lane setwise. Lemma 3.1 gives their common unsigned internal rule:

$$\begin{aligned} C^{(j)} \xrightarrow{r} C^{(h)} &\iff h \equiv j + r \pmod{g}, \quad h \neq 0, \\ C^{(j)} \xrightarrow{u} \partial &\iff j + u \equiv 0 \pmod{g}. \end{aligned} \quad (5.2)$$

Thus every enabled label, disabled label, target lane, and boundary exponent agrees. For a_+ every edge sign is positive, so its exit-sign set is $\{+1\}$. For a_- every ordinary-lane internal and boundary edge is negative. A direct boundary edge from the source gives negative exit sign. The globally enabled label-zero return a_- , followed by the same boundary exponent, gives positive exit sign. These are actual paths from the same source, so

$$\text{ExitOri}_{a_+}(C_{\text{src}}) = \{+1\}, \quad \text{ExitOri}_{a_-}(C_{\text{src}}) = \{+1, -1\}. \quad (5.3)$$

Theorem 4.3 supplies the prescribed-pair witnesses, giving

$$\begin{aligned} \mathfrak{S}_{a_+}(\lambda_0; F) &= \{s_F^+\}, \\ \mathfrak{S}_{a_-}(\lambda_0; F) &= \{s_F^+, s_F^-\} \quad (F \text{ consecutive}). \end{aligned} \quad (5.4)$$

Both pair spectra are the five source edges by Corollary 4.4, whereas the two survivor injections in (5.4) are distinct. This proves the matched all- g obstruction.

Thus no rule depending only on this unsigned labelled graph and source pair data can reconstruct the survivor frontier throughout the declared regime. The theorem establishes insufficiency of the unsigned observable; it does not claim that a sign bit is a universally minimal invariant.

6 Scope and Open Boundaries

The proofs use two specific hypotheses. First, the source fills a five-point ordinary lane, while the punctured lane has capacity four. Without complete lane occupancy, the uniform support guard and capacity isolation no longer give the present graph. Second, ordinary-lane branch restrictions preserve or reverse cycle order. General stabilizer restrictions can produce cyclic orders other than the two carried by the signed lift.

The declared theorem does not classify those larger state spaces. Nor does it classify the terminal phase fiber for arbitrary formal placements: phase saturation within an ordinary lane is not an all-hole phase-collapse theorem across lanes. The terminal construction here retains the actual boundary exponent and its selected reachable representative.

There is also a length boundary. Realizing a phase may use multiple globally enabled returns. The signed graph determines unbudgeted raw reachability and survivor incidence, not shortest words, a credit inequality, or an admissible settlement schedule.

Finally, normalized raw incidence is not typed transfer membership. Authorization, ancestry, fixed transfer bindings, and typed handoffs are not fields of the present injection system. Any raw-to-typed bridge must supply its own same-witness theorem under existing semantics. Neither the positive frontier theorem nor the unsigned obstruction resolves Projectable-Origin Supply, recursive compatibility, or reset bounds.

7 Computational Artifacts

The evidence package is under `experiments/paper36/` in the [RIME repository](#). Paths below are relative to this directory.

Surface	Location	Role
Control	<code>signed_lane_audit.py</code>	exhaustive $g = 2, 3$ branch and source-case checks
Result	audit JSON	bounded source-addressed survivor controls
Validation	<code>validation/validate_package.py</code>	source binding and separately implemented finite replay

These finite controls check consistency in the declared domains; they do not prove the all- g theorems. The theorem statements and proofs in this manuscript remain authoritative.

8 Claim Status and Boundary

Claim	Status
Proposition 2.1: exact terminal pullback	algebraic proof for the declared witness type
Lemmas 3.1–3.2: uniform guard and phase saturation	data-independent proofs for all $g \geq 2$
Theorem 3.3: signed phase descent	exact reachable transition quotient in the full ordinary-lane local-dihedral regime
Lemmas 4.1–4.2: boundary lifting and rigidity	actual prescribed-pair witness and unique incidence per sign
Theorem 4.3 / Corollary 4.4: complete survivor frontier	both inclusions, disjointness, and pair spectrum proved
Theorem 5.1: unsigned non-descent	explicit matched family for every $g \geq 2$
Full stabilizer, partial lane occupancy, length budgets	not classified
Typed transfer, projectability, recursion, settlement	outside the theorem surface

9 Conclusion

In the declared full ordinary-lane regime, a reachable injection can be compressed to its lane and source-relative orientation. Uniform edge guards permit an entire graph path to lift from one concrete representative, and actual phase control realizes every prescribed consecutive pair at a fixed boundary edge. The reachable exit signs then determine the placements of all three survivors.

The matched reflection family shows why the unsigned graph is too coarse: it preserves all labelled lane mobility and pair-hitting data but changes the complete survivor frontier. Phase can be removed from the compressed reachability state; orientation cannot be discarded from this observable without losing incidence information. These conclusions remain raw, multi-lane, hypothesis-explicit, and independent of recursive or reset-bound claims.

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